

Based on my comprehensive research and analysis of the CMInDS (Centre for Machine Intelligence and Data Science) faculty database and their recent publications, I can now provide a detailed overview of their current research activities.

Comprehensive Research Analysis: Centre for Machine Intelligence and Data Science (CMInDS) - IIT Bombay - Current and Recent Research Activities

Based on extensive analysis of faculty profiles, recent publications, and current research projects spanning 2021-2025, the Centre for Machine Intelligence and Data Science at IIT Bombay represents India's pioneering interdisciplinary research center focused on transforming society through cutting-edge artificial intelligence and data science research.

Core Faculty Research Focus Areas

Prof. D. Manjunath (Head, CMInDS) leads groundbreaking research in data science and AI applications in power systems, IoT data analytics, decision making through control theory, optimization and game theory, mathematics and statistics foundations, machine learning and AI algorithms, social networks, and network science. His recent breakthrough work includes packet-pair dispersion analysis using discrete-time queueing theoretic approaches, in-network computation in random wireless networks with PAC methodologies achieving constant refresh rates and lower energy costs, load balancing via random local search in closed and open systems (which won Best Paper Award at ACM SIGMETRICS 2010), and comprehensive stochastic systems performance modeling. His cutting-edge contributions span distributed computation in sensor networks, distributed optimization for network resource provisioning, random geometric graph models for wireless networks, stochastic coverage processes, queueing system models for wireless networks, self-tuning mechanisms for optimal resource allocation in computer systems, and advanced Internet protocols including pricing, quality of service, and traffic engineering. [scholar.google+3](#)

Prof. Ankur A. Kulkarni (Kelkar Family Chair in Quantitative Finance) specializes in systems theory with strategic agents and informational complexities, conducting fundamental research on game theory, information theory, stochastic control, and operations research. His recent revolutionary work focuses on security, stealth, nudging, information disclosure and elicitation, particularly in finance and security domains. His breakthrough contributions include "Beyond Common Randomness: Quantum Resources in Decentralized Control," "Fraud Detection and Deterrence in Electronic Voting Machines: A Game-Theoretic Approach," and

"Classifier Design for Decentralised Sensing with Digital Communication". His cutting-edge research encompasses Bayesian persuasion frameworks for epidemic containment on heterogeneous networks, Shannon meets Myerson information extraction from strategic senders, minimax theorems for finite blocklength lossy joint source-channel coding over arbitrarily varying channels, and strategy-proof spectrum allocation among multiple operators in wireless networks. As a member of the IT-Project Advisory Board of SEBI and Research Advisor to Tata Consultancy Services, his work directly impacts Indian financial markets and algorithmic trading regulation. [CMinDS.pdf](#)[fiitb+4](#)

Prof. Parthe Pandit (Thakur Family Chair Assistant Professor) conducts pioneering research in computational and statistical aspects of machine learning, focusing on kernel machines as viable alternatives to deep neural networks. His revolutionary AI2050 Early Career Fellowship research seeks to revitalize kernel machines in the deep learning era through data-dependent supervised kernels and fast scalable training algorithms for kernel machines. His recent breakthrough work includes momentum accelerated training of kernel machines, achieving linear-time complexity for any kernel choice, any number of centers, and any sample size, with delightfully parallel GPU utilization reaching full resource efficiency compared to deep networks' typical 20-25% GPU memory utilization. His cutting-edge contributions span learning and inference in high dimensions, generalization behavior of interpolating machine learning models, scalable algorithms for training kernel machines, and geometric analysis of optimization landscapes in deep neural networks. [parthe.github+3](#)[youtube](#)

Prof. Arjun Bhagoji (Assistant Professor) focuses on robust and reliable machine learning, data scarcity challenges, and model auditing. His cutting-edge research explores how machine learning models behave when faced with unexpected or adversarial inputs, defining clear limits on model reliability and robustness, building more interpretable and robust AI systems for real-world applications, and training reliable AI models in privacy-preserving manner when data is decentralized. His recent breakthrough work includes lower bounds for robust multi-class classifiers accepted as Spotlight at NeurIPS 2023, automated censorship detection accepted at KDD 2023, diffusion models adapted to generate protocol-compliant network data for SIGMETRICS 2024, and model versioning techniques to create model sequences robust to transferable adversarial examples. His innovative contributions span content moderation policy analysis across top websites, unsupervised machine learning models discovering new blocking signatures for DNS censorship, drift mitigation in cellular networks, and auditing tools for social media content moderation systems. [linkedin+3](#)

Prof. Pranay Sharma (Assistant Professor) specializes in distributed learning systems, stochastic optimization, nonconvex optimization, min-max optimization, and reinforcement learning. His research focuses on building efficient distributed learning systems and understanding fundamental aspects of how machine learning models learn, including provable limits to learning, efficient online learning, scalable distributed learning, and simultaneous optimization of multiple objectives. His breakthrough contributions include model sparsity simplifying machine unlearning with 78 citations, STEM stochastic two-sided momentum algorithm achieving near-optimal sample and communication complexities for federated learning with 77 citations, federated reinforcement learning achieving linear speedup under Markovian

sampling with 75 citations, and federated minimax optimization with improved convergence analyses and algorithms. His cutting-edge work encompasses FedVARP tackling variance due to partial client participation in federated learning, communication network topology inference via transfer entropy, distributed online convex optimization with sublinear dynamic regret, and parallel restarted SPIDER for communication-efficient distributed nonconvex optimization.[linkedin+3](#)

Operational and Strategic Leadership

Dr. Khushaal Popli (Chief Operating Officer) brings over 12 years of academia and industrial experience in data science, artificial intelligence, and digital transformation. His expertise spans process control engineering, advanced analytics for mineral processing industries, machine learning-based virtual sensors, real-time process monitoring, and optimization using first-principle and data-driven models. His recent contributions include managing data science-based innovation and technology development for global processing operations, developing multi-scale fundamental models for flotation processes, soft sensors for online recovery and grade measurements using image processing and machine learning, state estimation using extended Kalman filters, and diagnostics for different faults and feed disturbances.[linkedin+2](#)

Taha Khot (Program Manager) manages operations of IIT Bombay's AI research center, developing workflows for center processes, strategy development, and organizational management. His work includes developing strategic frameworks for AI research translation and coordinating interdisciplinary research initiatives across the center.[linkedin+1](#)